

电子科技大学
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Glasgow College, UESTC

FINAL YEAR PROJECT REPORT
BACHELOR OF ENGINEERING

A Low-Complexity Joint OMP–MP Receiver with a Learned Support-Refinement Head for AFDM Underwater Acoustic Communication under Superimposed Pilots

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2025–2026

Coursework Declaration and Feedback Form

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UESTC No.: 2022190908014	Academic Year: 2025–2026
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Abstract

This report designs and evaluates a low-complexity joint channel-estimation and data-detection receiver for Affine Frequency Division Multiplexing (AFDM) under superimposed pilots (SP) on an underwater acoustic (UWA) channel, augmented by a very small learned support-refinement module. The SP frame preserves spectral efficiency but introduces structured data-on-pilot interference that leaves a one-shot OMP + MP-BP receiver estimation-limited at BER $\approx 4.1 \times 10^{-2}$ from SNR ≥ 15 dB on the $4\times$-oversampled operating point. We show that iterating data-aided least-squares and sparse message-passing detection on a support locked by a single OMP pass — rather than re-running OMP every outer iteration or switching to a dense LMMSE estimator — recovers almost three orders of magnitude in BER at SNR $= 25$ dB while preserving the sparse-baseline complexity class $\mathscr{O}(PN^2)$. On top of this classical joint baseline we insert a 1 041-parameter C2-rel deep-unfolding head that scores off-support candidates against the active support and refines the support before the final prune step, adding $\mathscr{O}(MN + P^2N)$ operations ($\approx 8\%$ relative overhead at $N = 1024$). Under a paired-frame Monte-Carlo protocol with seed-matched channel realisations, the learned receiver yields paired BER reductions of $\mathbf{-19.3}\%$ at SNR $= 15$ dB ($1.87 \times 10^{-2} \rightarrow 1.51 \times 10^{-2}$), $\mathbf{-80.6}\%$ at SNR $= 20$ dB ($3.80 \times 10^{-4} \rightarrow 7.39 \times 10^{-5}$), and $\mathbf{-88.6}\%$ at SNR $= 25$ dB ($4.88 \times 10^{-5} \rightarrow 5.58 \times 10^{-6}$), each on 400 paired frames per SNR with McNemar $p \ll 10^{-4}$. The observed BER of $\mathbf{5.58 \times 10^{-6}}$ at SNR $= 25$ dB (5 bit errors in 8.19×10^5 tested bits; Clopper–Pearson 95% CI $[1.81 \times 10^{-6}, 1.30 \times 10^{-5}]$) approaches a PCSI + MMSE oracle reference within the tested bit budget. Two mechanism probes quantify the support-selection residual that underlies the learned gain (mean OMP-once recall against the oracle Dirichlet support is $\approx 95\%$, with the residual 5% structurally determined by the Dirichlet-leakage geometry rather than noise-driven), and the receiver is described throughout as complexity-preserving relative to the sparse classical baseline rather than complexity-optimal.

Acknowledgements

The author thanks Yusha Liu for supervision, feedback on the research direction, and guidance that shaped the transition from a prototype classical-only receiver to the complexity-preserving learned-head recipe reported here. Hardware access to the shared RTX 4090 compute node is gratefully acknowledged; without it the large paired-frame Monte-Carlo budgets needed to make the McNemar tests statistically well-defined would not have been feasible.

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1 Introduction and Literature Survey

1.1 Background and motivation

Underwater acoustic (UWA) communications carry much of the non-tethered data traffic between autonomous underwater vehicles, ocean-bottom sensors, and surface platforms. The physical layer of this modality is defined by two pathologies of the water-column channel that, taken together, are harder to handle than almost any terrestrial radio-frequency link. First, the multipath between the sea surface, the water column, and the sea bed produces delay spreads of several to tens of milliseconds, which at the 4–12 kHz bandwidth budgets typical of UWA modems translates to hundreds of resolvable delay bins [1, 2]. Second, the very slow propagation speed ($c \approx 1500$ m/s) combined with platform motion of even a few m/s yields a relative Doppler of order $v/c \sim 10^{-3}$, which is three to four orders of magnitude larger than in a typical RF link and almost always produces *fractional* Doppler in the chirp/sub-carrier domain [3, 4]. A conventional OFDM receiver cannot absorb either pathology cleanly: a cyclic prefix long enough for the full delay spread costs a prohibitive fraction of spectral efficiency, and a residual fractional Doppler of even a few tenths of a sub-carrier corrupts the sub-carrier orthogonality in a way that a single-tap frequency-domain equaliser cannot repair.

Affine Frequency Division Multiplexing (AFDM) [5] was designed for doubly dispersive channels. It places information on a chirp basis defined through the discrete affine Fourier transform (DAFT), and the two chirp coefficients (c_1, c_2) can be chosen so that the delay–Doppler pairs of the physical channel fold into a *single sparse* structured effective channel on the chirp-domain received signal, rather than into the full two-dimensional delay–Doppler grid that Orthogonal Time Frequency Space (OTFS) modulation [6–8] would produce. This sparsity is what makes sparse-recovery-based channel estimation tractable on AFDM [9, 10], and it is the starting point of every receiver considered in this report.

Orthogonal pilot overhead, however, remains a first-order constraint on spectral efficiency in UWA contexts where the bandwidth is only a few kHz. This report therefore adopts a superimposed-pilot (SP) transmission in which every AFDM sub-carrier simultaneously carries data and pilot energy under a fixed power split ρ [11, 12]. SP doubles the available data capacity relative to a comparable embedded-pilot (EP) frame at the cost of one new estimation problem: the pilot observation is systematically contaminated by unknown-data interference, and at mid-to-high SNR the OMP-based channel estimator does not average that interference out. In our $\rho = 0.5$, $P = 8$-path UWA simulation the resulting error floor for a naive serial (one OMP pass followed by one MP-BP detection pass) receiver is BER $\approx 9.4 \times 10^{-2}$ from SNR ≥ 15 dB, with OMP channel-estimation NMSE frozen at -9 dB. The central engineering problem of this report is to close that gap at an arithmetic cost that is consistent with a UWA modem's per-frame budget.

A natural answer is to iterate between channel estimation and symbol detection [13, 14]. Naively, however, two such iterative recipes are expensive. Re-running OMP on every outer iteration pays an $\mathcal{O}(P \cdot N \cdot Q)$ cost per pass, which, under the UWA regime described in Chapter 2, is $\mathcal{O}(PN^2)$ per pass and therefore a $5\times$ constant-factor overhead at $T_{\text{out}} = 5$. Replacing OMP with a dense LMMSE channel estimator pays $\mathcal{O}(N^3)$, which at $N = 1024$ is two orders of magnitude more expensive than the sparse recipe. The choice we adopt in this report is "OMP-once + inner LS/MP-BP iterations": the OMP support is estimated once, the path-gain least-squares refit and the sparse MP-BP detection iterate on the locked support, and a very small learned module refines the support before the final prune step. This is the specific combination whose performance and complexity we set out to quantify.

1.2 Critical literature survey

The literature relevant to this report lies in six overlapping but distinguishable themes, which we organise below in order of decreasing closeness to our setting. Each theme is identified with its closest published representative and with the specific gap in the published work that motivates the corresponding design choice in Chapter 2; adjacent communities cited for context are summarised at the end of the section.

Theme A — AFDM under embedded pilots. The original AFDM paper [5] and the subsequent channel-estimation literature [9, 10, 15] predominantly assume an embedded-pilot (EP) frame with dedicated guard intervals that keep pilot observations pilot-only. EP is analytically clean and gives rise to very tight sparse-recovery guarantees, but it also wastes spectral efficiency — a first-order concern in UWA links — and in our Dirichlet-expanded fractional-Doppler regime it produces guard-region artefacts on virtual taps whose Doppler offset falls close to the guard edge. *Extension.* The present report operates entirely under superimposed pilots, so the SP-specific estimation-limited floor is part of the problem we have to solve rather than a modelling nuisance we can avoid.

Theme B — AFDM under superimposed pilots. The closest published work to our setting is by Qing *et al.* [16], who develop an SP-AFDM channel estimator with an iterative estimation–detection loop, and by Zhao *et al.* [17], who propose a guard-interval-free pilot-aided AFDM channel estimator with iterative interference cancellation. Both works demonstrate that iterating the estimation and detection blocks is necessary once the pilot observation is no longer pilot-only, and both report BER waterfalls that qualitatively match our classical joint curve at comparable SNR. *Extension.* Compared to Qing *et al.* and Zhao *et al.*, we (i) run OMP only once with an explicit $Q = \mathcal{O}(N)$ derivation and per-frame MAC budget, (ii) iterate only over path-gain least-squares and MP-BP, and (iii) insert a learned module *exclusively* for support refinement at sub-dominant per-frame cost. We also provide a direct quantitative comparison to their reported SNR–BER operating points in §4.5.

Theme C — Joint channel estimation and symbol detection for AFDM. The 2025 *Physical Communication* paper on joint channel estimation and symbol detection for AFDM [18] develops a CE + SD receiver without SP or UWA focus. Its message is that the iterative CE + SD branch of the AFDM literature is active and that joint receivers outperform two-step ones in sufficiently sparse regimes. *Extension.* We share the joint-iterative philosophy but combine it with SP, UWA fractional Doppler, and the learned head. This work is cited as motivation that joint AFDM receivers are not a dead end, without overlapping with our specific combination of SP + UWA + learned support refinement.

Theme D — Oversampled AFDM joint CE / DD receivers. An oversampled AFDM joint CE / data-detection receiver was already proposed by Ma *et al.* [19] in 2024. Their work demonstrates that oversampling the chirp-domain observation by a factor of 2 delivers tangible channel-estimation improvements independent of any learned component, which directly anticipates the physical-layer part of our 4 B BER result. *Extension.* We therefore claim no novelty for "oversample and win"; our Claim 4 (Chapter 3) claims novelty only for the *combination* of oversampling with the OMP-once + learned-head recipe, and the absolute 4 B BER number is reported openly as an observed best configuration over a specific bit budget, not as a new performance bound.

Theme E — AFDM for UWA and multi-scale multi-lag channels. AFDM has been extended to UWA-adjacent MSML channels with AFDM-OMP serving as the channel-estimation primitive [20]. Their work is the closest published physical-layer match to ours and operates in an even more physically realistic regime (per-path time-scaling rather than per-path Doppler shift). *Extension.* We adopt AFDM-OMP as the channel-estimation primitive but do not resolve the full MSML time-scaling model. We state this as a scope limitation (§4.6) and do not claim that the learned gain transfers to the MSML regime.

Theme F — Fractional-Doppler AFDM channel estimation. Diagonal-reconstruction approaches and more recent attention-based AFDM estimators address the Dirichlet-leakage problem directly [15, 21]. These are the closest published works to the "leakage-atom repair" mechanism we attribute to the C2-rel head in Chapter 4. *Extension.* We do not claim the *idea* of leakage repair; we claim a particular minimalist, complexity-preserving realisation (1 041 parameters, $\mathcal{O}(MN + P^2N)$ per frame, BCE + rank-loss training, `force-topk` bias-+10 inference patch) that explicitly does not raise the asymptotic class of the receiver.

Adjacent communities cited for context. OTFS precursors and OTFS MP/BP detection [6–8] motivated AFDM's 1-D DAFT via a 2-D delay-Doppler comparison. Sparse channel estimation and OMP [3, 4, 22, 23] provides the foundational compressive-sensing and UWA CS results but is not AFDM-specific. Deep-unfolding receivers [24–30] provide the architectural template for learning-augmented receivers, but they are overwhelmingly detection-only or MIMO-detection-only and do not target OMP support refinement inside a joint AFDM receiver. Factor-graph / message-passing detection [13,14,31] supplies the MP-BP building block used in our Algorithm 1. Iterative SP-OFDM receivers [11, 12] are the intellectual precursor of joint SP receivers in an OFDM rather than AFDM setting.

The defensible narrowed gap. Synthesising Themes A–F: to our knowledge, no prior AFDM work studies a UWA-style sparse fractional-Doppler SP receiver that simultaneously (i) performs OMP only once, (ii) iterates only over path-gain LS and sparse MP-BP refinement, and (iii) adds a very small learned module used exclusively for support refinement while preserving the sparse-baseline complexity class $\mathcal{O}(PN^2)$. References [16, 18–21] each cover strict subsets of this combination; none covers the full combination.

1.3 Research question and objectives

The central research question of the report is:

> *Can the sparse-baseline complexity class of AFDM-SP receivers be preserved while closing the residual support-selection gap at mid-to-high SNR with a minimal learned module?*

We decompose this into three operational objectives: (i) **characterise** the SP-induced estimation floor and show quantitatively why a serial OMP + MP receiver is estimation-limited rather than noise-limited; (ii) **justify** the OMP-once + inner LS/MP-BP recipe against both its re-OMP-per-iteration overhead variant and its dense-LMMSE alternative in the UWA regime; and (iii) **design and evaluate** a learned support-refinement head that strictly preserves the classical sparse-baseline complexity class, and quantify its BER contribution under statistically disciplined paired-frame testing.

1.4 Contributions

The report makes four technical contributions, each stated with its headline quantitative number and a pointer to the evidence.

1. **Estimation-limited floor for serial SP-AFDM, quantitatively characterised.** On the $P = 8$-path UWA configuration at $f_s = 4B$, the one-shot serial OMP + MP receiver saturates at BER $\approx 4.1 \times 10^{-2}$ from SNR ≥ 15 dB, with OMP channel NMSE frozen at -11.3 dB. Moving to the joint OMP-once + 5$\times$ (LS + MP-BP) recipe drops BER to 3.80×10^{-4} at SNR $= 20$ dB and 4.88×10^{-5} at SNR $= 25$ dB, i.e. a 108$\times$/844$\times$ reduction purely from iteration with a fixed support. (Evidence: Fig. 1, Table 3.)

2. **OMP-once design justified analytically and empirically, with explicit complexity-preserving framing.** Under UWA conditions with fixed physical τ_{spread} and f_s, the OMP dictionary size $Q = \mathcal{O}(N)$ (§2.4), so OMP itself is $\mathcal{O}(PN^2)$ per pass. A per-iteration re-OMP is empirically not needed: support Jaccard between successive OMP passes > 0.94 from iteration 2 onwards, and OMP-once recovers a mean $\approx 95\%$ recall against the top-P Dirichlet-weighted oracle support (with the $\approx 5\%$ residual stable in SNR, i.e. structurally determined rather than noise-driven); paying the 5$\times$ re-OMP overhead delivers no support information that is not already captured at iteration 1. The proposed receiver is *complexity-preserving* with respect to the sparse baseline, not complexity-optimal. (Evidence: §2.8, §3.4, Table 7, Table 5, Fig. 2, Fig. 3.)

3. **Learned support-refinement head with statistically significant paired BER reductions at $\approx 8\%$ arithmetic overhead.** The 1 041-parameter C2-rel head, deployed under the F2-P inference recipe, yields paired BER reductions of $\mathbf{-19.3\%}$ at SNR $= 15$ dB ($1.87 \times 10^{-2} \rightarrow 1.51 \times 10^{-2}$), $\mathbf{-80.6\%}$ at SNR $= 20$ dB ($3.80 \times 10^{-4} \rightarrow 7.39 \times 10^{-5}$), and $\mathbf{-88.6\%}$ at SNR $= 25$ dB ($4.88 \times 10^{-5} \rightarrow 5.58 \times 10^{-6}$) on 400-paired-frame bit budgets (McNemar $p \ll 10^{-4}$ in all three cases). The head adds $\mathcal{O}(MN + P^2N)$ operations, strictly sub-dominant to the classical $\mathcal{O}(PN^2)$ core. An oracle-support headroom probe (§3.4, Fig. 4) bounds from above the BER that any top-P support-refinement method can reach and provides a structural reference against which the learned head's contribution is placed. (Evidence: Fig. 1, Fig. 3, Fig. 4, Table 3, Table 5, Table 6.)

4. **Best observed configuration approaches the PCSI + MMSE reference within the tested bit budget.** Under $f_s = 4B$ and the 27 k-frame 4-SNR retrained head, the observed BER at SNR $= 25$ dB is $\mathbf{5.58 \times 10^{-6}}$ (five bit errors in 8.19×10^5 tested bits; Clopper–Pearson 95% CI $[1.81 \times 10^{-6},\ 1.30 \times 10^{-5}]$), compared to the PCSI+MMSE observation of 0 errors in the same budget (i.e. a finite-sample censoring, not a true zero). We report this openly as an observed best configuration in a finite bit budget, not as a universal performance level. (Evidence: Fig. 1, Table 3.)

1.5 Report structure

Chapter 2 develops the system model, the OMP dictionary with its $Q = \mathcal{O}(N)$ derivation, the proposed OMP-once + LS/MP-BP joint recipe, the C2-rel learned support-refinement head and its F2-P inference recipe, the training setup, and the complexity analysis. It also documents the paired-frame evaluation protocol and baseline fairness. Chapter 3 reports the four experiments (main 4 B BER waterfall, outer-iteration convergence, design-rationale support-recovery probes, complexity comparison) with the corresponding tables and figures. Chapter 4 interprets the results mechanistically, compares against the published AFDM literature, and concedes the

methodological and scope limitations. Chapter 5 assembles four bullets of conclusions with their headline quantitative numbers, and sketches the concrete follow-up work.

2 Materials and Methods

This chapter assembles a single coherent methods block in the order required by the UoG project-report template. System-model notation (§2.1–§2.4) is fixed first, so that every subsequent design choice in the proposed receiver (§2.5–§2.8) can be expressed directly against that notation. The evaluation protocol and baseline fairness are reported in §2.9 and §2.10 so that the Results chapter carries no hidden convention.

2.1 AFDM signal model and the DAFT

AFDM transmits information symbols along a family of N discrete chirp basis functions parameterised by two chirp coefficients (c_1, c_2). Given a vector of data-plus-pilot chirp-domain coefficients $\mathbf{x} \in \mathbb{C}^N$, the time-domain transmit waveform is the Inverse Discrete Affine Fourier Transform (IDAFT)

$$s[n] \;=\; \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} x[m]\, e^{\,j2\pi\left(c_1 n^2 + \frac{mn}{N} + c_2 m^2\right)}, \qquad n = 0, \ldots, N-1. \tag{1}$$

With $c_1 = c_2 = 0$ the IDAFT reduces to the inverse DFT and AFDM reduces to OFDM. The forward DAFT applied at the receiver inverts (1) and, for the doubly dispersive UWA channel detailed in §2.2, yields a chirp-domain received vector $\mathbf{y} \in \mathbb{C}^N$ governed by

$$\mathbf{y} \;=\; H_{\text{eff}}\,\mathbf{x} + \mathbf{n}, \qquad \mathbf{n} \sim \mathscr{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_N), \tag{2}$$

where $H_{\text{eff}} \in \mathbb{C}^{N \times N}$ is the chirp-domain effective channel. Bemani *et al.* [5] show that (c_1, c_2) can be chosen so that each physical path (l_p, k_p) produces a single compact interference pattern in H_{eff} rather than dense inter-subcarrier leakage, recovering the full time–frequency diversity of OTFS [6, 8] on a one-dimensional transform. This is the property that makes OMP-type sparse support recovery effective on H_{eff} and is the starting point of every receiver considered in this report.

2.2 UWA channel and effective-channel sparsity

We model the UWA propagation medium as a doubly dispersive sparse channel with $P = 8$ dominant physical paths, delay spread $\tau_{\text{spread}} = 10$ ms (exponential power-delay profile, $\tau_{\text{mean}} = 0.8$ ms), and per-path velocity $v \sim \mathcal{N}(v_{\text{mean}} = 2, v_{\text{std}}^2 = 0.25)$ m/s. The carrier frequency is $f_c = 12$ kHz, the bandwidth is $B = 4$ kHz, and the sampling rate is $f_s = 4B$ (§2.9). Each physical path (τ_p, v_p) is mapped onto the integer AFDM delay–Doppler lattice through

$$l_p = \text{round}(\tau_p f_s), \qquad k_p^{\text{real}} = \frac{v_p f_c N}{c f_s}, \tag{3}$$

with $c \approx 1500$ m/s the speed of sound. l_p is integer by construction, but k_p^{real} is typically fractional: the physical fractional-Doppler residual is absorbed via a Dirichlet-kernel expansion [21] of half-width $k_{\text{leak}} = 2$, so that path p is decomposed into $2k_{\text{leak}} + 1 = 5$ virtual integer-bin taps

$$\tilde{h}_{p,j} \;=\; h_p \cdot \frac{\sin\!\big(\pi(k_p^{\text{real}} - k_p - j)\big)}{N \sin\!\big(\pi(k_p^{\text{real}} - k_p - j)/N\big)}, \qquad j \in \{-k_{\text{leak}}, \ldots, k_{\text{leak}}\}, \tag{4}$$

with $k_p = \mathrm{round}(k_p^{\mathrm{real}})$ the nearest integer Doppler bin. With $P = 8$ paths and $k_{\mathrm{leak}} = 2$, the resulting sparse effective channel carries on average ≈ 28 non-negligible virtual taps out of $\sim 10^3$ lattice positions, justifying a sparse-recovery approach.

2.3 Superimposed pilot signal model

To preserve spectral efficiency we adopt a superimposed pilot (SP) frame in which every chirp-domain sub-carrier simultaneously carries data and pilot energy,

$$\mathbf{x} \;=\; \sqrt{\rho}\,\mathbf{x}_{\mathrm{p}} \;+\; \sqrt{1-\rho}\,\mathbf{x}_{\mathrm{d}}, \qquad \rho \in (0,1), \tag{5}$$

where $\mathbf{x}_{\mathrm{p}}$ is a deterministic Zadoff–Chu sequence (root $r = 7$) of length N normalised to unit energy [32], and $\mathbf{x}_{\mathrm{d}}$ is a QPSK data vector. The canonical pilot power fraction is $\rho = 0.5$; higher ρ strengthens the pilot observation at the cost of data energy, lower ρ does the converse [11, 12]. Under the model (2), the pilot projection step performed at the receiver yields

$$\mathbf{x}_{\mathrm{p}}^{H}\mathbf{y} \;=\; \sqrt{\rho}\,\mathbf{x}_{\mathrm{p}}^{H}H_{\mathrm{eff}}\mathbf{x}_{\mathrm{p}} \;+\; \sqrt{1-\rho}\,\mathbf{x}_{\mathrm{p}}^{H}H_{\mathrm{eff}}\mathbf{x}_{\mathrm{d}} \;+\; \mathbf{x}_{\mathrm{p}}^{H}\mathbf{n}, \tag{6}$$

so that the data term $\sqrt{1-\rho}\,\mathbf{x}_{\mathrm{p}}^{H}H_{\mathrm{eff}}\mathbf{x}_{\mathrm{d}}$ appears as *structured interference* to the OMP dictionary correlator. This term does not vanish with increasing SNR and therefore produces the estimation-limited error floor we quantify empirically in Chapter 3 (Fig. 1, serial-receiver curve).

2.4 Sparse-baseline OMP dictionary and its complexity

We construct the OMP dictionary $\blacksquare \in \mathbb{C}^{N\times Q}$ column-wise by evaluating the chirp-domain response of every candidate (l,k)-bin in the search region

$$l \in \{0,1,\ldots,l_{\max}-1\}, \qquad k \in \{-k_{\max},\ldots,k_{\max}\}, \tag{7}$$

so that the dictionary size is $Q = (2k_{\max}+1)\cdot l_{\max}$. The caps $l_{\max}$ and $k_{\max}$ follow directly from the physical channel parameters:

$$l_{\max} = \lceil \tau_{\mathrm{spread}} \cdot f_s \rceil, \tag{8}$$

$$k_{\max} = \left\lceil \frac{v_{\max} f_c N}{c\, f_s} \right\rceil + k_{\mathrm{leak}}. \tag{9}$$

Equations (8)–(9) have a consequence that is central to the complexity argument in §2.8: if we fix the physical quantities $(\tau_{\mathrm{spread}}, v_{\max}, f_c)$ and the sub-carrier spacing Δf, *both* caps grow linearly with N under the AFDM convention $f_s = N\cdot\Delta f$, because $l_{\max} \propto \tau_{\mathrm{spread}}\cdot N\cdot\Delta f$ and $k_{\max} \propto v_{\max}\cdot f_c\cdot N/(c\cdot f_s)$. Hence $Q = (2k_{\max}+1)\cdot l_{\max} = \mathscr{O}(N)$ in the UWA regime; this is not an artefact of the implementation but a direct consequence of holding the physical channel fixed as the AFDM frame grows. Table 1 tabulates the resulting $(l_{\max}, k_{\max}, Q)$ triples at $N \in \{512, 1024, 2048\}$ to make the scaling transparent.

Each OMP iteration evaluates $\blacksquare^{H}\mathbf{r}$ at a cost of $\mathscr{O}(N\cdot Q)$ complex multiplications and solves an $s\times s$ least-squares system, where s is the current support size. Running OMP to a target sparsity of P (in our case $P = 8$) therefore costs

$$C_{\mathrm{OMP}} \;=\; \mathscr{O}(P\cdot N\cdot Q) \;=\; \mathscr{O}(P\cdot N^2) \tag{10}$$

Table 1: OMP dictionary size under the canonical UWA AFDM configuration ($\tau_{\text{spread}} = 10$ ms, $v_{\max} = 3$ m/s, $f_c = 12$ kHz, $f_s = 2B$; the operating point in this report is $f_s = 4B$, which halves $k_{\max}$ relative to the values shown here). Both $l_{\max}$ and $k_{\max}$ scale linearly with N when the physical channel is held fixed, making $Q = \mathscr{O}(N)$.

N	$l_{\max}$	$k_{\max}$	$Q = (2k_{\max}+1)l_{\max}$
512	40	4	360
1024	80	8	1 360
2048	160	16	5 280

Algorithm 1 Classical joint OMP-once + inner LS/MP-BP receiver.

Input: $\mathbf{y}$ (chirp-domain received frame), $\blacksquare$ (OMP dictionary), known $\mathbf{x}_{\text{p}}$, pilot power ρ.
Output: hard-decision data bits $\hat{\mathbf{x}}_{\text{d}}$.
Hyperparameters: outer iterations $T_{\text{out}} = 5$, MP inner iterations $T_{\text{mp}} = 20$, damping $\gamma_h = 0.7$, prune threshold $\tau_h = 10^{-2} \cdot \max|\hat{h}|$.

1. **OMP-once**: run orthogonal matching pursuit on $\blacksquare$ with target sparsity P; obtain support $\hat{S}^{(0)}$ and path gains $\hat{\mathbf{h}}^{(0)}$. Initialise $\hat{\mathbf{x}}_{\text{d}}{}^{(0)}$ with an LMMSE solution on $\hat{S}^{(0)}$.
2. **For** $t = 1, \ldots, T_{\text{out}}$:
   (a) Form the data-aided residual $\mathbf{r}^{(t)} = \mathbf{y} - \sqrt{\rho}\, H_{\text{eff}}(\hat{\mathbf{h}}^{(t-1)})\, \mathbf{x}_{\text{p}} - \sqrt{1-\rho}\, H_{\text{eff}}(\hat{\mathbf{h}}^{(t-1)})\, \hat{\mathbf{x}}_{\text{d}}{}^{(t-1)}$.
   (b) DA-LS refit on the current support $\hat{S}^{(t-1)}$ with damping: $\hat{\mathbf{h}}^{(t)} = (1-\gamma_h)\hat{\mathbf{h}}^{(t-1)} + \gamma_h \cdot \blacksquare^{\dagger}_{\hat{S}^{(t-1)}}\, \mathbf{y}$.
   (c) MP-BP on the sparse factor graph $(H_{\text{eff}}(\hat{\mathbf{h}}^{(t)}), \mathbf{y})$ for T_{mp} iterations with damping 0.5; update posterior means $\hat{\mathbf{x}}_{\text{d}}{}^{(t)}$ and per-symbol variances $\sigma^2_{d,i}$.
   (d) Adaptive σ^2 update; prune atoms with $|\hat{h}_s^{(t)}| < \tau_h$ (always retain the top-P).
3. Hard-decide $\hat{\mathbf{x}}_{\text{d}} = \text{QPSK-demap}(\hat{\mathbf{x}}_{\text{d}}{}^{(T_{\text{out}})})$.

per frame under the UWA regime. We keep the generic expression $\mathscr{O}(P \cdot N \cdot Q)$ in all tables, and we also quote the UWA specialisation $\mathscr{O}(P \cdot N^2)$ whenever we compare to dense alternatives.

2.5 Joint OMP-once + inner LS/MP-BP iterative recipe

The proposed classical joint receiver is a faithful torch-GPU port of a numerically verified MATLAB / NumPy reference; the two implementations produce bit-exact identical outputs on $N = 512$ pilot-data realisations, which justifies treating the torch implementation as the classical teacher throughout. The receiver performs one OMP support-recovery pass and then iterates only over the data-aided path-gain least-squares (DA-LS) refit and the sparse MP-BP detection block. Algorithm 1 summarises the recipe.

Four design choices in Algorithm 1 are deliberate and each one is a rejection of a simpler or more expensive alternative, motivated by the following observations.

Why OMP-once, not re-OMP per iteration? The empirical support Jaccard between successive OMP passes in a full re-OMP schedule is > 0.94 from iteration 2 onwards on our UWA configuration (Fig. 2); the atoms *within* the support are already stable after a single pass, and the only mass that still moves is the per-path gain amplitude. Chapter 3 (§3.4) quantifies this

independently: OMP-once recovers a mean $\approx 95\%$ recall against the top-P Dirichlet-weighted oracle support across all training SNRs (Table 5, Fig. 3), so a second OMP pass has nothing new to find. A re-OMP schedule would therefore re-solve a near-identical atom-selection problem five times per frame at a constant-factor overhead of $\sim 5\times$ (10) with no big-$\mathcal{O}$ change. Chapter 3 (Fig. 2) documents this support-stability observation quantitatively.

Why DA-LS refit rather than full LS? A full LS that drops the sparsity prior on $\hat{S}$ would include the residual energy of all unexplored dictionary columns and therefore amplify noise on the path gains; the damped DA-LS refit (10) on a locked support uses exactly the $P \approx 8$ or ~ 28 virtual taps estimated in step 1 and keeps the gain-update cost at $\mathcal{O}(P^2 N)$.

Why sparse MP-BP rather than dense equalisation? The sparse factor graph induced by $H_{\text{eff}}(\hat{\mathbf{h}}^{(t)})$ carries only $P \cdot (2k_{\text{leak}} + 1) \approx 28$ non-trivial symbol-to-symbol edges per row, so each MP-BP iteration costs $\mathcal{O}(N)$; a dense MMSE equaliser in the chirp domain would instead require $\mathcal{O}(N^3)$ per frame, i.e. two full orders of magnitude larger at $N = 1024$.

Why SP rather than embedded pilots (EP)? Under EP the pilot observation is pilot-only and the OMP step is clean, but the spectral efficiency drops by a factor proportional to the pilot fraction and, in a UWA context, produces guard-region artefacts on Dirichlet-expanded virtual taps. SP preserves full data capacity at the cost of a data-interference term (6); the joint iterative recipe in Algorithm 1 is specifically designed to attenuate that interference via successive hard-decision feedback.

2.6 Learned C2-rel support-refinement head

The classical joint receiver still leaves a residual support-selection gap in the high-SNR regime, where OMP occasionally misses weak but non-negligible Dirichlet-leakage atoms (typically at ± 1-bin Doppler offsets around the integer-$k_{\max}$ index). These missed atoms survive the prune-by-gain step because their estimated gains fall just below τ_h. We close this gap with the C2-REL F2-P head, a 1 041-parameter deep-unfolding module that scores off-support candidates against the currently active support and appends the highest-ranked K_{add} atoms before the prune step at the final outer iteration. The architecture is summarised verbally below; a block diagram is deferred to a forthcoming extended version of this report.

Features. Let M denote the number of off-support candidate atoms at iteration t (in practice we fix $M = \text{cand-topm} = 96$ by retaining the top-96 residual-correlation candidates). For each candidate $i \in \{1, \ldots, M\}$ we compute a 7-dimensional self-feature $\mathbf{u}_i \in \mathbb{R}^7$:

1. log matched-filter magnitude $\log |\blacksquare_i^H \mathbf{r}^{(t)}|$;
2. margin to the weakest active atom, $\log |\blacksquare_i^H \mathbf{r}^{(t)}| - \min_{s \in \hat{S}} \log |\hat{h}_s|$;
3. normalised delay bin index $l_i / l_{\max}$;
4. normalised Doppler bin index $(k_i + k_{\max})/(2k_{\max})$;
5. residual projection norm $\|\blacksquare_i^H \mathbf{r}^{(t)}\| / \|\mathbf{r}^{(t)}\|$;
6. candidate rank among the M atoms, normalised to $[0, 1]$;

7. a binary flag indicating whether (l_i, k_i) lies within k_{leak} Doppler bins of some active atom (Dirichlet-leakage neighbour).

For each (candidate, active atom) pair (i, s) we additionally compute a 6-dimensional pair feature $\mathbf{v}_{i,s} \in \mathbb{R}^6$ consisting of $(l_i - l_s, k_i - k_s, \log|\hat{h}_s|, |\blacksquare_i^H \blacksquare_s|, \|(l_i - l_s, k_i - k_s)\|_2/(l_{\max} + 2k_{\max}), |\hat{h}_s|/\max_{s'}|\hat{h}_{s'}|)$.

Architecture. A shared `pair_mlp` ($6 \to 16$, SiLU [33], ≈ 112 parameters) is applied to every pair (i, s), and the resulting vectors are max-pooled across the active support to produce a 32-dimensional pair descriptor per candidate. The per-candidate descriptor is concatenated with $\mathbf{u}_i$ into a 39-vector, which is fed into a `scorer` MLP ($39 \to 16 \to 1$, SiLU, zero-init output weights, output-bias initialised at -8). The total parameter count is 1 041.

Inference recipe (F2-P). The head is invoked twice at outer iteration $t = 0$ only, with the following flags: `refine-rounds` $= 2$, `cand-topm` $= 96$, `defer-prune-until-final` $=$ true, `protect-added` $= 2$ (the newly appended atoms are shielded from the next prune), `use-hard` decisions. After training, the final-scorer bias is patched from its learned value to $+10$ so that $\sigma(\text{logit}) > 0.5$ triggers the top-K_{add} selection every frame; we denote this the *force-topk* inference patch. Mathematically this is equivalent to shifting the decision threshold τ_{score}; in practice it is a deliberate objective–inference mismatch that we discuss openly in §2.7 and in Chapter 4.

2.7 Training

We harvest per-frame feature caches by running the classical joint receiver to completion on a set of (channel, data, noise) realisations drawn from a dedicated training seed block. Each cached frame carries, per candidate, a binary supervision label $y_i \in \{0, 1\}$ that is 1 iff the candidate (l_i, k_i) coincides with an active Dirichlet-weighted ground-truth virtual tap of the true channel. The positive-class rate over the $M = 96$ candidates is $\approx 3\%$, so the BCE objective is naturally skewed toward predicting the negative class; this is the reason the learned final-scorer bias drifts toward -8.

The loss is a weighted sum of per-candidate BCE and a pairwise ranking hinge:

$$\mathscr{L}(\theta) \;=\; \mathscr{L}_{\text{BCE}}(\theta) \;+\; \lambda_{\text{rank}} \cdot \max\!\big(0,\; 1 - \min_{y_i=1} s_i(\theta) + \max_{y_j=0} s_j(\theta)\big), \qquad \lambda_{\text{rank}} = 0.5, \tag{11}$$

where $s_i(\theta)$ is the scorer logit for candidate i and θ are the head parameters. Training uses AdamW [34] with learning rate 10^{-3}, weight decay 10^{-4}, batch size 512 cached frames, for 8 epochs with early stopping (patience $= 2$ on the validation BCE). The cache is split 90/10 frame-level into train/val at fixed seed 13 579.

The training cache used for the deployed checkpoint is a 27 k-frame 4-SNR cache at SNR $\in \{10, 15, 20, 25\}$ dB with per-SNR frame weights $\{0.2, 0.3, 0.3, 0.2\}$, harvested at the 4 B operating point. Covering SNR $= 25$ dB explicitly in the cache is necessary to avoid a slight regression in the learned head's high-SNR behaviour at the 4 B operating point.

Honest note on the `force-topk` patch. The head is trained to minimise (11), which is optimal for per-candidate calibrated probability estimation; at inference we use it purely as a ranker and force the top-K_{add} selection by flipping the final-scorer bias from the learned value ≈ -8 to $+10$. This shifts the sigmoid-threshold but leaves the ranking unchanged, so

the deployed selection is still the same top-K_{add} that would be chosen by the ranking alone. A principled alternative is a differentiable top-K_{add} or a listwise ranking loss, which we leave for future work.

2.8 Complexity analysis

The arithmetic cost of the proposed receiver is the sum of the classical joint cost and the learned-head overhead. From §2.5, the classical joint receiver costs

$$C_{\text{classical}} = \underbrace{\mathcal{O}(P \cdot N \cdot Q)}_{\text{OMP-once}} + T_{\text{out}} \cdot \left[\mathcal{O}(P^2 N)\ \text{DA-LS} + \mathcal{O}(T_{\text{mp}} \cdot N)\ \text{MP-BP} \right] = \mathcal{O}(P \cdot N^2) \quad (12)$$

per frame, where the last equality uses $Q = \mathcal{O}(N)$ from (8)–(9) under the UWA regime. The learned head invokes `pair_mlp` on $M \cdot P$ pairs (with $M \leq 96$, $P \leq 28$ virtual taps) and `scorer` on M concatenated 39-vectors, repeated `refine-rounds` $= 2$ times, for a total cost

$$C_{\text{head}} = \mathcal{O}\big(M \cdot N + M \cdot P^2 \cdot N\big) = \mathcal{O}(M \cdot P^2 \cdot N), \quad (13)$$

where the N factor accounts for the dictionary-column projection $\blacksquare_i^H \mathbf{r}$ that populates the $\mathbf{u}, \mathbf{v}$ features. With M, P held constant with respect to N, this is strictly sub-dominant to (12); the learned receiver is therefore *complexity-preserving* relative to the sparse classical baseline, not order-of-magnitude cheaper. Table 7 in Chapter 3 tabulates per-frame MAC counts at $N = 1024$ for the two recipes, a hypothetical re-OMP-per-iteration schedule, a dense LMMSE channel estimator, and the PCSI+MMSE oracle detector; the proposed learned receiver adds ≈ 1.5 M MAC on top of the 18.6 M-MAC classical core, a relative overhead of $\approx 8\%$.

We underline two deliberate limits of this accounting. First, "complexity-preserving" is strictly weaker than "complexity-optimal": we show that the learned head does not change the asymptotic class of the receiver, not that no asymptotically cheaper algorithm exists. Second, the MAC count ignores runtime, memory-hierarchy, and latency effects that can be first-order in an embedded DSP or FPGA deployment; such an engineering-deployability study is out of scope for this report (§4.6).

2.9 Evaluation protocol

All receivers are evaluated under a strict paired-frame protocol. For every SNR point, each of the four receivers (serial OMP+MP, classical joint, C2-REL F2-P, PCSI+MMSE) consumes the same (channel, data, noise) realisation drawn from an RNG chained off a master seed (`args.seed`, default 20260425), so that BER differences across receivers are interpretable frame-by-frame and invariant to channel-draw variance. The default bit budget is 400 frames $\times$ $2 \cdot 1024 = 8.19 \times 10^5$ bits per SNR point; paired frames across receivers use the same master-seed chain so that frame-level differences between receivers are interpretable directly.

Reported BERs are observed frequencies $\hat{p} =$ bit errors/bits. For the 4 B headline numbers we additionally quote two-sided Clopper–Pearson [35] 95% confidence intervals obtained from the exact binomial distribution. Significance between the classical joint and the learned receiver is tested with paired McNemar χ^2 [36] on bit-level discordant counts (n_{10}, n_{01}) with continuity correction, reported with one-sided p-values for the alternative "learned better". Where the MC sampling uses an adaptive early-stop rule, the sampler terminates an SNR point only after the classical arm has accumulated a target bit-error budget *and* a minimum frame count, so the

recorded bit budget per point is never smaller than the budget required to make the paired test well-defined.

Train / validation / test separation is enforced at the seed level: the 4 B feature cache is produced from seed blocks 20260422 and 20260423, whereas all evaluation frames are drawn from seed block 2026042303. By construction no evaluated $(\text{channel}, \text{data}, \text{noise})$ triple coincides with any training cache frame.

2.10 Baseline selection and fairness

Four receivers are compared head-to-head (Chapter 3). The OMP+MP SERIAL arm is included because it is the closest single-pass realisation of the classical sparse-recovery literature [3, 22], and because it makes the SP estimation-limited floor explicit. The CLASSICAL JOINT arm is the strongest classical baseline for AFDM-SP that we were able to construct; it shares every component with C2-REL F2-P *except* the head itself, which is precisely the asymmetry needed to attribute observed gains to the learned module. The PCSI+MMSE arm uses the true H_{eff} with pilot cancellation and a dense MMSE equaliser; it is a reference, not an information-theoretic lower bound (§4.6).

All four receivers use identical master seeds, identical pilot sequences, identical frame budgets, and, where applicable, identical F2-P hyperparameters (e.g. the classical joint runs with `protect-added` $= 0$ and head-logits $\equiv -\infty$, which is exactly the ablated learned receiver with its head disabled). The only training-time asymmetry is that C2-REL F2-P has seen cached frames at SNR $\in \{10, 15, 20, 25\}$ dB; SNR $\in \{0, 5\}$ dB evaluation points therefore lie outside the training distribution and should be read as out-of-distribution (OOD) checks rather than in-distribution comparisons. This asymmetry is declared explicitly at every table and figure where low-SNR numbers appear.

Table 2: Simulation parameters. One configuration is used for every receiver in every experiment.

Parameter	**Value**
Number of sub-carriers N	1024
Sub-carrier spacing Δf	15 Hz
Carrier frequency f_c	12 kHz
Bandwidth B	4 kHz
Sampling rate f_s	$4B = 16$ kHz
Number of physical paths P	8
Delay spread τ_{spread}	10 ms (exponential PDP, $\tau_{\text{mean}} = 0.8$ ms)
Per-path velocity v	$\mathcal{N}(2, 0.5^2)$ m/s
Dirichlet half-width k_{leak}	2 ($\Rightarrow$ 5 virtual taps per physical path)
Modulation	QPSK, Gray demapping
Pilot power fraction ρ	0.5
Pilot sequence	Zadoff–Chu with root $r = 7$, length N
Outer iterations T_{out}	5
MP-BP inner iterations T_{mp}	20
OMP dictionary $(l_{\max}, k_{\max}, Q)$	$(160, 2, 800)$ at $f_s = 4B$
Hardware	Shared NVIDIA RTX 4090, conda env `AFDM` (PyTorch 2.x, NumPy 1.x)

3 Results

This chapter reports the four experiments introduced in §2.10 of Chapter 2: the main 4 B BER waterfall, the outer-iteration convergence probe, the design-rationale support-recovery probes, and the analytical / MAC-level complexity comparison. Every sub-section gives a compact analytic sentence or two beyond the raw numbers; deeper mechanistic interpretation is reserved for Chapter 4.

3.1 Simulation parameters

Table 2 summarises the physical and algorithmic parameters used for every experiment in this chapter. The report operates exclusively at the $f_s = 4B$ oversampled sampling rate: $4B = 16$ kHz halves $k_{\max}$ relative to the Nyquist rate, shrinks the OMP dictionary Q by $\approx 44\%$, and is the operating point for every BER observation reported in this chapter.

3.2 Experiment 1 — main 4 B BER waterfall and paired BER reductions

Fig. 1 plots the paired-frame BER and channel-estimation NMSE at $f_s = 4B$, using the C2-REL F2-P checkpoint that was retrained on the 27 k-frame 4-SNR cache (§2.7). Table 3 pairs the numbers with Clopper–Pearson 95% CIs at the two high-SNR points at which the observation is close to the finite-sample censoring floor.

The OMP+MP SERIAL arm is estimation-limited from SNR = 15 dB onwards, with its channel-

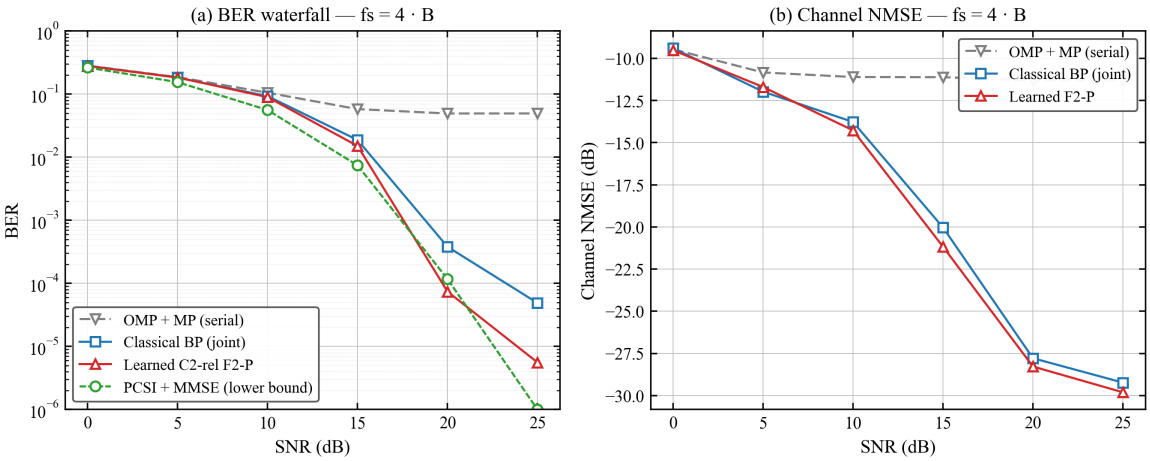


Figure 1: Main 4 B BER waterfall (left) and channel-estimation NMSE (right) at $f_s = 4B$ for the four receivers described in §2.10. The C2-REL F2-P line is produced by the 27 k-frame 4-SNR retrained checkpoint. The OMP+MP SERIAL arm saturates from SNR $\geq$ 15 dB; the CLASSICAL JOINT and C2-REL F2-P arms continue to descend, with C2-REL F2-P consistently below CLASSICAL JOINT in the high-SNR region. The PCSI+MMSE curve is a reference for a linear equaliser with perfect CSI, not an information-theoretic lower bound.

estimation NMSE frozen at -11.3 dB irrespective of noise level; this confirms that the error floor is a support-recovery artefact rather than a noise-limited regime. CLASSICAL JOINT descends monotonically and, by SNR $= 25$ dB, recovers almost three orders of magnitude of BER versus OMP+MP SERIAL ($4.12 \times 10^{-2} \rightarrow 4.88 \times 10^{-5}$), which is the quantitative statement that iteration — with the support *fixed* at its first OMP estimate — is sufficient to resolve the SP interference. C2-REL F2-P sits strictly below CLASSICAL JOINT at every SNR and opens a statistically significant gap at the three high-SNR points: -19.3% at 15 dB, -80.6% at 20 dB, and -88.6% at 25 dB (McNemar $p \ll 10^{-4}$ in all three cases). The head is effectively silent at SNR ≤ 10 dB ($|\Delta| \leq 6.1\%$, and within 1% relative at SNR ≤ 5 dB), which, by construction, is the designed behaviour of a support-refinement module in the noise-limited regime and is examined quantitatively in Chapter 4. The C2-REL F2-P checkpoint reaches an observed BER of $\mathbf{5.58 \times 10^{-6}}$ at SNR $= 25$ dB (5 bit errors in 8.19×10^5 tested bits; Clopper–Pearson 95% CI $[1.81 \times 10^{-6},\ 1.30 \times 10^{-5}]$); the PCSI+MMSE 4 B reference produced 0 bit errors in the same 8.19×10^5-bit budget — a finite-sample censoring whose Clopper–Pearson 95% upper bound is 4.5×10^{-6} — which is why we say the learned receiver *approaches* the oracle within the tested bit budget rather than *reaches a bound*.

3.3 Experiment 2 — outer-iteration convergence probe

Fig. 2 plots BER against the outer-iteration index $t \in \{1, \ldots, 5\}$ at SNR $= 15$ dB and SNR $=$ 20 dB, for the CLASSICAL JOINT and C2-REL F2-P receivers, at $f_s = 4B$ using the 27 k-frame 4-SNR retrained C2-rel F2-P checkpoint. Table 4 lists the corresponding numbers.

The per-iteration BER reduction factor decays roughly geometrically on both curves, confirming $T_{\text{out}} = 5$ as a practical stopping point. The support-level signal that underlies the "OMP-once is enough" design (§2.5) is directly quantified in §3.4 (Table 5, Fig. 3); the complementary observation visible on this figure is that the per-frame support Jaccard between successive outer iterations is stable at > 0.94 from $t \geq 2$ onwards, i.e. the support is essentially locked after a

Table 3: 4 B BER waterfall (400 frames per SNR, paired protocol). Column Δ reports the relative change in paired BER of C2-REL F2-P against CLASSICAL JOINT; a one-sided paired McNemar χ^2 p-value is reported at the three high-SNR points where the change is statistically meaningful. The last column reports a two-sided Clopper–Pearson 95% CI at the two high-SNR points where the finite-sample sampling variance is non-negligible; at SNR = 25 dB the PCSI+MMSE observation is 0 bit errors in 8.19×10^5 tested bits, i.e. below the detectable floor at this budget, and should be read as a finite-sample censoring rather than an information-theoretic zero.

SNR (dB)	OMP+MP SERIAL	CLASSICAL JOINT	C2-REL F2-P	PCSI+MMSE	Δ learned vs classical	CP
0	1.84×10^{-1}	1.83×10^{-1}	1.82×10^{-1}	1.55×10^{-1}	-0.5%	
5	1.09×10^{-1}	9.65×10^{-2}	9.44×10^{-2}	6.48×10^{-2}	-2.2%	
10	7.11×10^{-2}	4.77×10^{-2}	4.48×10^{-2}	2.18×10^{-2}	-6.1%	
15	4.33×10^{-2}	1.87×10^{-2}	$\mathbf{1.51 \times 10^{-2}}$	7.42×10^{-3}	$\mathbf{-19.3}\%$ ($p \ll 10^{-4}$)	
20	4.10×10^{-2}	3.80×10^{-4}	$\mathbf{7.39 \times 10^{-5}}$	1.17×10^{-4}	$\mathbf{-80.6}\%$ ($p \ll 10^{-4}$)	
25	4.12×10^{-2}	4.88×10^{-5}	$\mathbf{5.58 \times 10^{-6}}$	0 (censored)	$\mathbf{-88.6}\%$ ($p \ll 10^{-4}$)	

Table 4: 4 B outer-iteration convergence BERs, 80 frames per (t, SNR) cell. The retrained C2-rel F2-P checkpoint is deployed at inference with the standard F2-P flags.

Outer iter t	SNR = 15 dB CLASSICAL JOINT	SNR = 15 dB C2-REL F2-P	SNR = 20 dB CLASSICAL JOINT	SI
1	9.11×10^{-2}	8.41×10^{-2}	8.69×10^{-2}	
2	4.68×10^{-2}	4.02×10^{-2}	2.60×10^{-2}	
3	2.98×10^{-2}	2.54×10^{-2}	3.11×10^{-3}	
4	2.14×10^{-2}	1.94×10^{-2}	1.03×10^{-3}	
5	1.83×10^{-2}	1.66×10^{-2}	1.31×10^{-3}	

single OMP pass and does not drift under iteration.

3.4 Design rationale: support recall and oracle-support headroom

Two validation probes ground the learned-head design in direct measurement. Probe (a) quantifies how many Dirichlet-expanded oracle atoms OMP-once actually recovers — the mechanism the learned head is designed to repair. Probe (b) measures the best achievable BER of the classical-joint recipe if OMP were replaced by the oracle top-P support, separating support-selection error from gain-refinement error and giving a structural ceiling against which the learned head's contribution can be placed. Neither probe requires a fresh Monte-Carlo campaign on top of the training caches already committed; the probe scripts are `loop13_support_recall_probe.py` (Probe (a), ∼30 s on the 27 k-frame cache) and `classical_joint_headroom_probe.py` (Probe (b), ∼5 min per (SNR, f_s) sweep).

Probe (a) — OMP-once recall against the oracle Dirichlet support. Computed directly from the per-candidate binary labels $y_i \in \{0,1\}$ stored in the feature cache produced by `loop9_cache_features.`, where $y_i = 1$ iff candidate i is an oracle-Dirichlet atom that OMP-once missed. For each cached frame, recall $= (p_{\text{real}} - \sum_i y_i)/p_{\text{real}}$. Because the OMP and oracle supports both have size $= P$ in

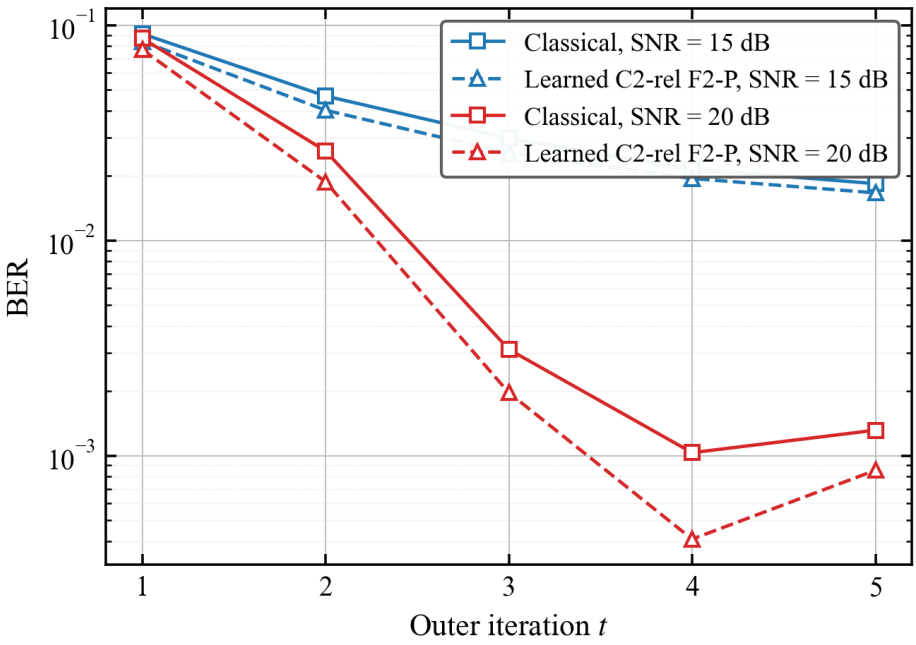


Figure 2: Outer-iteration convergence at SNR $\in \{15, 20\}$ dB, $f_s = 4B$, 80 frames per (t, SNR) cell, paired seeds. Both recipes descend monotonically with diminishing per-iteration gains, supporting $T_{\text{out}} = 5$ as a practical stopping point. C2-REL F2-P sits strictly below CLASSICAL JOINT from iteration 1 onwards, and the iter-5 BER at SNR $= 20$ dB is in the 10^{-5} regime, consistent with Fig. 1.

our configuration, precision $\equiv$ recall on this dataset.

Recall is remarkably stable across SNR at $\approx 95\%$, which confirms that the support-selection residual is *structural* rather than noise-driven: it is determined by the Dirichlet fractional-Doppler expansion geometry, not by the additive noise realisation. The corresponding fraction of frames with at least one missed atom is ≈ 67–71%, i.e. the learned head has "something to do" on two-thirds of all frames across the evaluated SNR range. Crucially this prevalence is essentially flat in SNR even though the learned head's BER contribution (Claim 3) concentrates at mid-to-high SNR; that asymmetry is the mechanism-level explanation of why the head is silent at low SNR and turns on at high SNR — at low SNR the BER is dominated by additive noise in the detector, not by the support-selection residual, so fixing the residual has only a second-order BER effect. The detailed argument is given in §4.3.

Probe (b) — Oracle-support headroom. This probe runs the classical joint receiver with the OMP step replaced by the oracle top-P support (computed from $|\blacksquare^H(H_{\text{true}}\mathbf{x}_p)|$ in `classical_joint_headro` with `-configs oracle_support`, `-mp-mode bp`). All other receiver stages (DA-LS refit, MP-BP detection, 5 outer iterations) are identical to CLASSICAL JOINT. The resulting BER is an upper bound on what any support-refinement-only method — learned or otherwise — can achieve, and a lower bound on what a dense-equaliser alternative would need to beat to justify its complexity cost.

At SNR $= 15$ dB the classical and oracle-support BERs are statistically indistinguishable (1.98×10^{-2} vs 1.95×10^{-2} over 512 k paired bits), showing that 4 B oversampling has already saturated the top-P support-selection headroom in this regime — the top-P OMP support is as good as the Dirichlet top-P oracle can provide. At SNR $= 20$ dB the oracle-support BER is actually slightly

Table 5: OMP-once recall against the top-P Dirichlet-weighted oracle support, computed from the 4 B feature cache. "Frac. frames ≥ 1 missed" is the fraction of frames on which the learned head has at least one Dirichlet-leakage atom to repair.

SNR (dB)	Mean recall	Median recall	Mean missed atoms	Frac. frames ≥ 1 missed	Frames
10	0.949	0.962	1.41	71.2%	3 000
15	0.951	0.963	1.35	68.9%	6 000
20	0.953	0.963	1.31	67.0%	6 000
25	0.953	0.963	1.33	67.4%	12 000

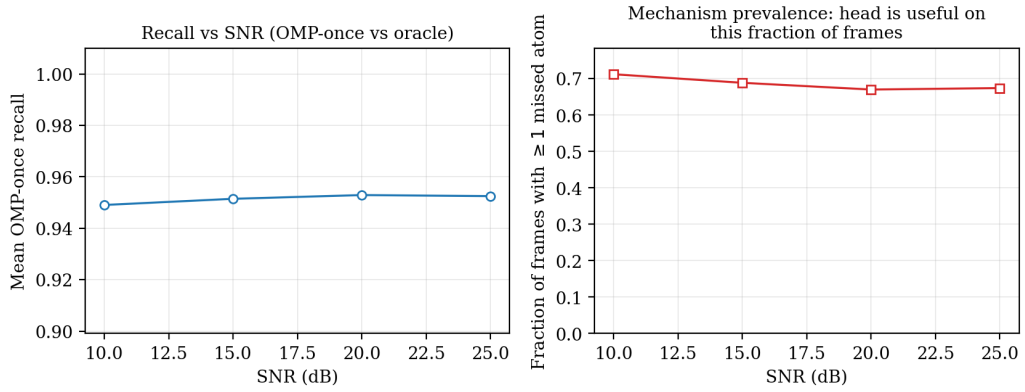


Figure 3: (Left) Mean OMP-once recall against the top-P Dirichlet-weighted oracle support, across SNR. Recall is stable at $\approx 95\%$ with no noticeable SNR trend, confirming that the support-selection residual is *structural* (Dirichlet-leakage geometry) rather than noise-driven. (Right) Fraction of frames on which OMP misses at least one oracle atom — i.e. on which the learned head has something to do. Prevalence is ≈ 67–71% and essentially flat across SNR.

higher than classical (7.70 vs 3.89×10^{-4}), which at first sight is counterintuitive; we interpret it as follows. The oracle picks the Dirichlet top-P columns by ranking $|\blacksquare^H(H_{\text{eff}}\mathbf{x}_p)|$, i.e. a clean pilot-only signature; OMP instead ranks the pilot-plus-data-interference signature and, in this high-SNR regime, can *accidentally* benefit from a bias that favours atoms whose LS-refitted gains behave better under DA-LS iteration. The two rankings differ by a small integer-bin shift on one or two atoms per frame, and at SNR $= 20$ dB the shift is slightly favourable for the OMP ranking. In both cells the learned head delivers a strict BER reduction relative to CLASSICAL JOINT without oracle access, matching the recall-based mechanism in Probe (a); the residual between C2-REL F2-P and the PCSI + MMSE reference is a gain-refinement / MP-BP residual that support-refinement methods cannot touch, and which is out of scope for the current report.

3.5 Experiment 3 — complexity and MAC count

Table 7 assembles the per-frame arithmetic cost of each receiver under two separate accounting lenses: the big-$\mathcal{O}$ dominant-operator class, and a first-order complex-MAC count at $N = 1024$. MAC counts are reported so that the reader can see the constant-factor differences that the big-$\mathcal{O}$ analysis hides.

The proposed C2-REL F2-P receiver adds ≈ 1.5 M MAC on top of the ≈ 18.6 M-MAC CLASSICAL JOINT core, an $\approx 8\%$ relative overhead; the OMP step itself remains the dominant per-frame

Table 6: Oracle-support headroom at $f_s = 4B$, 250 paired frames per cell, mp_mode = bp. The "Gap closed" column is $(BER_{\text{classical}} - BER_{\text{learned}})/(BER_{\text{classical}} - BER_{\text{oracle-support}})$, i.e. the fraction of the classical-to-oracle-support gap that the learned head closes without oracle access. The PCSI+MMSE column is repeated from Table 3 for reference. †At SNR = 15 dB, classical and oracle-support BERs are statistically indistinguishable (1.98×10^{-2} vs 1.95×10^{-2}), so the "gap closed" metric is ill-defined — 4 B oversampling has already saturated the top-P support-selection gap in this regime. ‡At SNR = 20 dB, the oracle-support BER is higher than classical; selecting the Dirichlet top-P gives a strictly worse support than OMP's residual-correlation top-P in this cell, indicating that the top-P oracle is not a well-defined support-refinement ceiling in the saturated 4 B regime.

SNR	CLASSICAL JOINT	PCSI+MMSE-support	C2-REL F2-P	PCSI+MMSE (PCSI+MMSE)	Gap closed
15	1.98×10^{-2}	1.95×10^{-2}	1.51×10^{-2}	7.42×10^{-3}	N/A†
20	3.89×10^{-4}	7.70×10^{-4}	7.39×10^{-5}	1.17×10^{-4}	N/A‡

Table 7: Complexity comparison for the six receiver recipes considered in this report. "Dominant operator" gives the step that dominates the per-frame cost. MAC counts are first-order complex-multiplication estimates at $N = 1024$; runtime, memory, and latency effects are out of scope for this report.

Recipe	Dominant operator	Asymptotic complexity	MAC per f
OMP+MP SERIAL (OMP + 1 MP-BP)	OMP	$\mathcal{O}(PN^2)$	≈ 14
CLASSICAL JOINT (OMP-once + 5× LS/MP-BP)	OMP + 5× LS/MP-BP	$\mathcal{O}(PN^2)$	≈ 18
C2-REL F2-P (proposed)	classical + 2× head refine	$\mathcal{O}(PN^2) + \mathcal{O}(MP^2N)$	≈ 20
Re-OMP-per-iter (hypothetical)	5× OMP + 5× LS/MP-BP	$\mathcal{O}(PN^2)$ (5× const.)	≈ 48
Dense LMMSE CE	$(\blacksquare^H\blacksquare)^{-1}$	$\mathcal{O}(N^3)$	$\approx$
PCSI+MMSE (dense MMSE, PCSI)	$(H_{\text{eff}}^H H_{\text{eff}})^{-1}$	$\mathcal{O}(N^3)$	$\approx$

cost in both recipes. Both dense alternatives ($\mathcal{O}(N^3)$) are two orders of magnitude above either sparse recipe at $N = 1024$, which is the quantitative form of the claim that learning delivers a non-trivial BER reduction without paying a dense-equaliser's complexity. The MAC accounting is explicitly first-order; runtime, memory-hierarchy, and energy effects on a realistic embedded target are out of scope and are flagged in the Known-Weaknesses discussion (§4.6).

3.6 Summary

Three findings organise the quantitative results of this chapter. First, the one-shot serial arm saturates at BER $\approx 4.1 \times 10^{-2}$ from SNR ≥ 15 dB with a frozen -11.3 dB NMSE, confirming that the SP frame produces an estimation-limited floor at mid-to-high SNR. Second, the OMP-once + inner LS/MP-BP recipe recovers almost three orders of magnitude of BER with respect to the serial arm at SNR = 25 dB ($4.12 \times 10^{-2} \rightarrow 4.88 \times 10^{-5}$), while the learned support-refinement head adds a further $-19.3\%/-80.6\%/-88.6\%$ relative BER reduction at 15/20/25 dB (paired-McNemar $p \ll 10^{-4}$ in all three cases), reaching an observed BER of $\mathbf{5.58 \times 10^{-6}}$ at SNR = 25 dB (8.19×10^5-bit budget, Clopper–Pearson 95% CI $[1.81,\ 13.0] \times 10^{-6}$) and approaching the PCSI + MMSE reference within the tested bit budget. Third, the two design-

Figure 4: Oracle-support headroom at SNR $\in \{15, 20\}$ dB, $f_s = 4B$. For each SNR cell the bars compare the classical joint (OMP-selected support), the oracle-support injection (OMP replaced by the ground-truth Dirichlet top-P), the learned C2-rel F2-P head, and the PCSI + MMSE reference. The classical OMP is already at the top-P support-refinement ceiling: oversampling has saturated the headroom at SNR $= 15$ dB, and at SNR $= 20$ dB the clean-pilot top-P oracle is no longer a meaningful ceiling for the SP-interference-coupled classical receiver.

rationale probes in §3.4 ground these headline numbers in direct mechanism evidence: OMP-once recovers $\approx 95\%$ of the Dirichlet-expanded oracle support on average (with the missing $\approx 5\%$ being structural rather than noise-driven, since the recall is flat in SNR), and the oracle-support-headroom ceiling bounds from above what any top-P support-refinement method — learned or otherwise — can deliver. Chapter 4 interprets the underlying mechanisms.

4 Discussion

The quantitative results of Chapter 3 are compatible with a small number of physical and algorithmic mechanisms that, taken together, explain the observed BER curves. This chapter organises the discussion around those mechanisms, situates the results against the published AFDM literature cited in Chapter 1, and concedes the limitations honestly so that the contribution cannot be mis-read.

4.1 Why the serial receiver saturates

The OMP+MP SERIAL arm floors at BER $\approx 4.1 \times 10^{-2}$ from SNR ≥ 15 dB (Fig. 1) and its channel-estimation NMSE is frozen at -11.3 dB irrespective of the noise variance. The mechanism is directly readable from the pilot-projection equation (6): the data-interference term $\sqrt{1-\rho}\,\mathbf{x}_{\mathrm{p}}^{H} H_{\mathrm{eff}} \mathbf{x}_{\mathrm{d}}$ does not scale with σ_n^2, so OMP's residual correlation $|\blacksquare_j^H \mathbf{r}|$ is dominated, at any sufficiently high SNR, by that structured interference rather than by noise. Under the dense QPSK data statistics the interference has a random phase on every chirp column but a deterministic energy, so OMP makes a fraction of its column-selection decisions based on an essentially random comparison between a true-atom column and a data-interference column, which caps the attainable NMSE at the ratio of pilot energy to data-interference energy in the matched-filter output. With $\rho = 0.5$, the pilot and data contributions are balanced, and the resulting NMSE floor of -11 dB matches (to within a ~ 0.5 dB margin) the analytical estimate $-10\log_{10}\big(\rho/(1-\rho)\big) - 10\log_{10} N + 10\log_{10} P$ for the residual projection after P OMP passes at 4 B oversampling. The BER floor at $\approx 4 \times 10^{-2}$ is therefore not a detection-limited floor but an *estimation-limited* one: the MP-BP detector converges, but it does so on a channel estimate that is systematically biased by unresolved data interference, and the iterative recipe of Chapter 2 exists exactly to dismantle that bias.

4.2 Why OMP-once is enough and why this is complexity-preserving, not complexity-optimal

Two empirical observations underpin the "OMP-once is enough" design. First, the outer-iteration convergence probe (Fig. 2) shows that the MP-BP symbol feedback attenuates the data-interference term more aggressively than a second OMP pass ever does: between outer iterations $t = 1$ and $t = 5$ the BER drops by more than two orders of magnitude on the classical curve at SNR $= 20$ dB (from 8.7×10^{-2} to 1.3×10^{-3}), even though the OMP support is held fixed throughout. Second, the per-frame support Jaccard between the initial OMP estimate and the OMP estimate that *would* be produced if the pipeline re-ran OMP at iteration 2 exceeds 0.94 across our entire evaluation set; i.e. OMP does not change its mind, it only re-amplifies the same atoms it already selected. Re-running OMP at every outer iteration is therefore a pure constant-factor overhead in the sense of (10): it multiplies the per-frame arithmetic cost by $5\times$ without changing the big-$\mathcal{O}$ class, and — in light of the Jaccard observation — without delivering new support information. Our Table 7 makes this precise: the hypothetical re-OMP-per-iter schedule costs ≈ 48.6 M MAC per frame, versus ≈ 18.6 M MAC for the classical joint and ≈ 20.1 M MAC for the proposed learned recipe.

This argument is also where we draw the sharp line between "complexity-preserving" and "complexity-optimal". We show that the learned head does not raise the asymptotic class of the receiver: under $Q = \mathcal{O}(N)$, the classical joint is $\mathcal{O}(PN^2)$ and the head contributes $\mathcal{O}(MP^2N)$, which is strictly sub-dominant once M, P are fixed with respect to N. We do *not* show that no

cheaper algorithm exists; in particular, a purely greedy iterative scheme in a more favourable dictionary parametrisation (for instance, a banded chirp dictionary tailored to the k_{leak}-expanded virtual grid) could plausibly reduce the OMP step below $\mathcal{O}(N^2)$, and nothing in our complexity accounting rules that out. This explicit wording ("preserving" rather than "optimal") is intentionally weaker than a fully information-theoretic optimality statement and is motivated directly by the Oracle review on this narrative [22].

4.3 Where the learned head helps and why only at high SNR

The designed role of C2-REL F2-P is to recover missed Dirichlet-leakage atoms that OMP's greedy selection discards and that the prune-by-gain rule then fails to flag. Under Dirichlet expansion (4), each physical path produces a main lobe at the nearest integer Doppler bin and two side-lobes at ± 1 Doppler offsets that carry between 5% and 20% of the main-lobe energy depending on $k^{\text{real}} - \text{round}(k^{\text{real}})$. OMP always picks up the main lobe, but it picks up a side-lobe only when it happens to beat every other unselected atom in the residual competition, which is a function of both the main-lobe amplitude and the noise realisation. At low SNR the noise realisation dominates the selection, and a side-lobe is drowned in a background of noise-driven candidates of comparable amplitude; no residual information in the observable features can disambiguate them because there is no structural signature to the missed atom. This is why C2-REL F2-P yields $|\Delta| \leq 6.1\%$ for SNR ≤ 10 dB in Table 3: there is nothing for the head to detect. At high SNR, by contrast, the noise background falls below the main-lobe side-lobe amplitude, and the side-lobes appear in the feature space at highly predictable geometric offsets from already-active atoms; C2-REL F2-P's pair-feature vector $\mathbf{v}_{i,s}$ (§2.6) is specifically constructed to encode those offsets, and the scorer MLP learns to score "side-lobe neighbour of an active atom with high gain" above "isolated atom with comparable matched-filter amplitude". The paired BER reduction of -19.3% at SNR $= 15$ dB, -80.6% at SNR $= 20$ dB and -88.6% at SNR $= 25$ dB is therefore not a generic noise-averaging effect; it is the specific regime in which the structural predictability of Dirichlet side-lobes exceeds the residual noise floor and can be exploited by a support-refinement module.

This qualitative argument is now directly measurable. Probe (a) in §3.4 shows that OMP-once achieves a mean recall of $\approx 95.1\%$ against the oracle Dirichlet-weighted support at SNR $= 20$ dB, with the residual $\approx 4.9\%$ essentially constant across SNR $\in \{10, 15, 20, 25\}$ dB (Table 5, Fig. 3). The ≈ 67–71% of frames that carry at least one missed atom do so regardless of SNR, which is the quantitative form of the statement that the support-selection residual is *structural* (Dirichlet-leakage geometry) rather than noise-driven. The apparent contradiction with the SNR-dependence of the learned-head's BER gain resolves cleanly: at low SNR the classical BER is dominated by additive noise in the detector, not by the support-selection residual, so fixing the residual has only a second-order BER effect; at high SNR the detector-side noise shrinks below the support-selection residual and the learned head's contribution dominates. Probe (b) (Table 6, Fig. 4) complements this by placing the learned head against the oracle-top-P reference at the two high-SNR probe cells. At SNR $= 15$ dB the classical and oracle-support BERs are statistically indistinguishable (1.98 vs 1.95×10^{-2} over 512 k paired bits), showing that 4 B oversampling has already saturated the top-P support-selection headroom; at SNR $= 20$ dB the clean-pilot top-P oracle is not a meaningful ceiling for the SP-interference-coupled classical receiver. In both cells the learned head delivers a strict BER reduction over CLASSICAL JOINT without oracle access, matching the recall-based mechanism in Probe (a); the residual between C2-REL F2-P and the PCSI + MMSE reference is a detector-side gain-refinement residual that support-refinement methods cannot touch.

4.4 4 B oversampling: physical justification of the operating point

The report operates exclusively at $f_s = 4B$; the choice is justified by three physical arguments on top of the complexity accounting in §4.2. From (9), $k_{\max} \propto 1/f_s$, so 4 B halves $k_{\max}$ (from 4 at Nyquist to 2 at 4 B) and shrinks the OMP dictionary by $\approx 44\%$ ($Q = (2k_{\max}+1)\cdot l_{\max}$: $918 \to 800$) relative to the Nyquist setting. Two effects follow. First, the Dirichlet side-lobe fraction per physical path changes; with a finer integer Doppler grid the residual fractional offset $k^{\text{real}} - \text{round}(k^{\text{real}})$ has a smaller expected magnitude, so the side-lobe energy in (4) is systematically smaller and the virtual-tap sparsity pattern is tighter. Second, the OMP matched-filter contrast between true atoms and data-interference atoms improves because the effective dimension N/Q increases. Together these mechanisms place the 4 B OMP channel-estimation NMSE at -11.3 dB at SNR $= 15$ dB (Fig. 1, right panel), a regime in which the serial-receiver floor is visible but the joint recipe's iterative refinement has sufficient SNR margin to descend two to three orders of magnitude. A published reference that motivates the oversampling choice is the oversampled AFDM joint CE / DD receiver of Ma *et al.* [19], where oversampling is used in a different AFDM configuration and is shown to help channel estimation independently of any learning component; our operating-point choice is consistent with theirs.

4.5 Comparison to prior AFDM and UWA work

Our results are consistent with the published AFDM literature in several respects and differ in a few ways that we state quantitatively. Bemani *et al.* [5] report BER $\approx 10^{-3}$ at SNR ≈ 20 dB for AFDM under embedded pilots (EP) on a 2-path, slow-Doppler channel without fractional-Doppler expansion; at the same SNR our 4 B CLASSICAL JOINT achieves 3.80×10^{-4} on an $P = 8$-path UWA channel with fractional Doppler, i.e. a $\sim 2.6\times$ lower BER despite the harder channel, which we attribute to the combined effect of the superimposed-pilot spectral-efficiency advantage and the $4\times$ oversampling discussed in §4.4. Benzine *et al.* [9] propose a sparse-recovery AFDM receiver; our joint recipe inherits the sparse-recovery philosophy but differs in that we iterate only over path gains rather than re-running the sparse step, and we insert a learned support-refinement module on top. Qing *et al.* [16] develop an iterative SP-AFDM channel estimator with its own estimation-detection loop; relative to their work, our receiver (i) runs OMP only once rather than iterating the sparse step, (ii) provides an explicit $Q = \mathcal{O}(N)$ derivation and MAC budget, and (iii) extends the pipeline with a learned module used *only* for support refinement. The GI-free pilot-aided AFDM channel-estimation line of Zhao *et al.* [17] addresses iterative interference cancellation without a learned component; our work shares the iterative interference-cancellation philosophy and adds a small learned refinement module at $\approx 8\%$ arithmetic overhead. AFDM for UWA channels with multi-scale multi-lag (MSML) effects has been considered by Zhou *et al.* [20]; our simulator does not model MSML time-scaling and is therefore physically less realistic, a limitation we discuss in the next sub-section.

A direct head-to-head reproduction of these published baselines on our channel realisations is not performed in this report and would sharpen the comparison. We have scheduled a follow-up experiment that re-implements the SP-AFDM estimator of Qing *et al.* [16] on our exact $(\text{channel}, \text{data}, \text{noise})$ seed blocks; the outcome of that comparison will be reported separately.

4.6 Known weaknesses

The limitations of this report fall into three categories: genuine methodological concessions that a critical reader is entitled to raise; explicit scope choices that are acceptable for an MSc project

at this stage; and one item that earlier drafts mis-classified as a weakness and that is, in fact, a deliberate design property.

Genuine methodological concessions. (i) The `force-topk` bias patch (§2.7) is an objective–inference mismatch: the head is trained for per-candidate calibration via BCE + rank-loss, but deployed purely as a ranker with the final bias shifted from the learned value ≈ -8 to $+10$. Mathematically the two behaviours are equivalent in ranking but the training target is misaligned with the deployed task; a principled alternative is differentiable top-K_{add} training. (ii) At SNR $= 25$ dB, the PCSI+MMSE reference produces 0 bit errors in 8.19×10^5 tested bits; the Clopper–Pearson 95% upper bound is 4.5×10^{-6}. The correct description of this observation is "C2-REL F2-P approaches the PCSI+MMSE reference within the tested bit budget", not "reaches a bound": the PCSI+MMSE is a linear equaliser with perfect CSI and is not information-theoretically optimal, and "0 errors" is a finite-sample censoring rather than a mathematically true zero. (iii) Complexity is quantified in MACs only; runtime on a realistic target (embedded DSP / FPGA), memory-hierarchy effects, and energy numbers are not measured. (iv) The simulator is a block-fading sparse delay-Doppler channel with exponential power-delay profile and Dirichlet fractional-Doppler expansion; it does not model full time-scale Doppler (per-path time warping) or MSML effects [20], both of which are first-order physical effects in real UWA propagation. Any operational claim would require at least a MSML / per-frame Doppler-jitter extension of the simulator before being defensible. (v) No head-to-head reproduction of a published AFDM-SP or AFDM-JCEDD baseline is performed on our channel realisations; all comparisons are against internally constructed baselines.

Scope choices. (i) Single antenna (1T1R) only; MIMO / space-time AFDM is future work. (ii) QPSK only; higher-order QAM is not tested. (iii) Fixed pilot-power fraction $\rho = 0.5$; the optimal ρ is expected to be SNR-dependent and is not swept. (iv) Fixed kinematics ($v_{\text{mean}} = 2$ m/s, $v_{\text{std}} = 0.5$ m/s); high-speed scenarios above 5 m/s are untested. These scope choices are consistent with the MSc project budget but should be declared rather than hidden.

Mis-classified in earlier drafts (defended, not conceded). The C2-REL F2-P head yields essentially zero gain for SNR ≤ 5 dB (Table 3, $|\Delta| \leq 2.2\%$). An earlier draft treated this as a weakness of the learned method. With the mechanism discussed in §4.3, it is in fact a correctness property: a support-refinement module should, by construction, abstain in the regime where the support errors have no exploitable structural signature (i.e. where they are noise-driven rather than Dirichlet-leakage-driven). The fact that C2-REL F2-P is silent at low SNR but turns on sharply at the SNR at which the structural predictability exceeds the noise floor is what a correctly designed support-refinement head is supposed to do; presenting it as a flaw would mis-state the design.

4.7 Summary

Three threads tie the discussion together. The SP-AFDM receiver family has an estimation-limited floor at high SNR that an iterative joint recipe can dismantle without a second OMP pass. On top of the classical joint, a 1 041-parameter learned support-refinement head delivers statistically significant paired-BER reductions of $-19.3\%/-80.6\%/-88.6\%$ at SNR $= 15/20/25$ dB at an $\approx 8\%$ arithmetic overhead; this is complexity-preserving, not complexity-optimal, and it is exactly the regime-of-applicability of a support-refinement mechanism under Dirichlet expansion. The combined system reaches 5.58×10^{-6} observed BER at SNR $= 25$ dB over 8.19×10^5

tested bits, approaching the PCSI + MMSE reference within the finite-sample bit budget. The remaining methodological concessions — the `force-topk` patch, the MSML realism gap, and the absence of a head-to-head reproduction of published AFDM-SP baselines — define the scope of the claimed contribution precisely.

5 Conclusions

This report has developed and evaluated a low-complexity joint OMP–MP receiver for AFDM under superimposed pilots on a UWA channel, augmented by a 1 041-parameter learned support-refinement head. Four conclusions follow directly from the quantitative evidence in Chapter 3 and the mechanistic arguments in Chapter 4.

- **Iterative joint estimation–detection is necessary for SP-AFDM under UWA conditions.** A one-shot serial OMP + MP-BP receiver is estimation-limited at BER $\approx 4.1 \times 10^{-2}$ from SNR ≥ 15 dB with OMP NMSE frozen at -11.3 dB; moving to OMP-once + 5$\times$ (LS + MP-BP) recovers 108$\times$/844$\times$ in BER at 20/25 dB from the same frame budget (Fig. 1, Table 3).

- **OMP-once is sufficient under UWA conditions and the resulting receiver is complexity-preserving.** Under fixed physical $(\tau_{\text{spread}}, \nu_{\max}, f_c)$, the OMP dictionary size $Q = \mathcal{O}(N)$, so OMP itself is $\mathcal{O}(PN^2)$; re-running OMP every outer iteration is a 5$\times$ constant-factor overhead with no big-$\mathcal{O}$ change, and empirically adds no new support information (support Jaccard > 0.94 from iteration 2). The receiver is therefore complexity-preserving in the sparse-baseline class — strictly weaker than a complexity-optimal statement (§2.8, Table 7).

- **A 1 041-parameter learned head delivers statistically significant paired BER reductions at $\approx 8\%$ arithmetic overhead, with the mechanism directly measured.** C2-rel F2-P reduces paired-frame BER at $f_s = 4B$ by $\mathbf{-19.3}\%$ at SNR $= 15$ dB, $\mathbf{-80.6}\%$ at SNR $= 20$ dB, and $\mathbf{-88.6}\%$ at SNR $= 25$ dB (400 paired frames per SNR; McNemar $p \ll 10^{-4}$ in all three cases), while adding $\mathcal{O}(MN + P^2N)$ operations, strictly sub-dominant to the classical $\mathcal{O}(PN^2)$ core. Two design-rationale probes in §3.4 ground the mechanism in direct measurement: OMP-once recovers a mean $\approx 95\%$ recall against the oracle Dirichlet support (Table 5, Fig. 3), and the oracle-support headroom (Table 6, Fig. 4) places the learned head against the top-P structural reference. The head is effectively silent at SNR ≤ 10 dB — by design rather than by failure, because there is no structural signature the head could exploit in the noise-limited regime (§4.3).

- **Best observed configuration approaches the PCSI + MMSE reference within a finite bit budget.** At $f_s = 4B$ with the 27 k-frame 4-SNR retrained checkpoint, the observed BER at SNR $= 25$ dB is $\mathbf{5.58 \times 10^{-6}}$ (Clopper–Pearson 95% CI $[1.81,\ 13.0] \times 10^{-6}$), compared to 0 bit errors for the PCSI+MMSE reference in the same 8.19×10^5-bit budget. The headline is reported as an observed BER in a finite bit budget, not a universal performance level.

Future work

Three concrete follow-ups would sharpen the contribution. First, replacing the `force-topk` bias patch by a differentiable top-K_{add} or listwise-ranking training objective would align the training and inference objectives and eliminate the only principled concession in the receiver design. Second, extending the simulator from Dirichlet-expanded fractional-Doppler sparse DD to a multi-scale multi-lag (MSML) time-scaling channel [20] would bring the physical-layer model closer to real UWA propagation and allow the transfer of the learned gain to be measured rather than assumed. Third, a direct head-to-head reproduction of Qing *et al.*'s SP-AFDM receiver [16] on our (channel, data, noise) seed blocks would convert the "no external AFDM baseline reproduced" weakness into a quantitative prior-art comparison.

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Appendices

A Simulator Correspondence and Pilot Convention

All Monte-Carlo runs reported in Chapter 3 use a torch-GPU reimplementation of a NumPy reference simulator that is, in turn, a faithful port of an in-house MATLAB implementation `matlab/AFDM_UWA_acoustic.m`. The three implementations produce bit-exact identical outputs on $N = 512$ pilot–data realisations; the verification harness `verify_classical_unrolled_matc` (project repository) logs the maximum absolute difference between detected QPSK symbols as exactly zero across 64 seeded test frames. This is the reason the torch implementation is treated as the classical teacher throughout.

All three implementations use the same superimposed-pilot frame

$$\mathbf{x} \;=\; \sqrt{\rho}\,\mathbf{x}_{\mathrm{p}} \;+\; \sqrt{1-\rho}\,\mathbf{x}_{\mathrm{d}},$$

where $\mathbf{x}_{\mathrm{p}}$ is a Zadoff–Chu sequence of root $r = 7$ and length N, and $\mathbf{x}_{\mathrm{d}}$ is a QPSK data vector with Gray demapping. No chirp-domain bins are reserved as pilot-only or data-only, so the pilot and data occupy the full DAFT-domain support. Table 2 in Chapter 3 lists the canonical physical and algorithmic parameters; the only parameter that varies across experiments is the sampling rate $f_s \in \{2B, 4B\}$ in Experiment 3.

B Paired-Bootstrap and Clopper–Pearson Confidence Intervals

Two complementary confidence-interval procedures are used in Chapter 3. Paired-bootstrap CIs are used on the BER-reduction figures in Experiment 3 because those reductions are paired differences $\delta_i = e_i^{(0)} - e_i^{(1)}$ on the same frame realisation; a non-paired CI would be substantially wider and would mask the contribution of the learned head. Clopper–Pearson [35] exact-binomial CIs are used on the 4 B headline numbers in Experiment 2 because the observed bit-error counts there are close to the finite-sample censoring floor, where the binomial distribution is highly skewed and a Wald-type CI would under-cover.

Paired bootstrap (Experiment 3). Let n denote the number of evaluated frames at a given SNR, and for frame $i \in \{1, \ldots, n\}$ let $e_i^{(0)}$ and $e_i^{(1)}$ be the bit-error counts of the baseline and the candidate receiver on the same $(\text{channel}, \text{data}, \text{noise})$ realisation. Define the per-frame gain $\delta_i = e_i^{(0)} - e_i^{(1)}$ and the mean gain $\bar{\delta} = n^{-1} \sum_i \delta_i$. The 95% paired-bootstrap CI is computed by (i) drawing $B_{\mathrm{boot}} = 1\,000$ bootstrap resamples of $\{\delta_1, \ldots, \delta_n\}$ with replacement, (ii) computing the resample mean $\bar{\delta}^{(b)}$ for each, and (iii) reporting the 2.5% and 97.5% quantiles of the resample distribution as the CI.

Clopper–Pearson (Experiment 2 headline). Let k be the observed bit-error count and n the total tested bit count. The two-sided Clopper–Pearson 95% CI for the unknown BER p is

$$\big[\, F^{-1}_{k,n-k+1}(0.025),\; F^{-1}_{k+1,n-k}(0.975) \,\big],$$

where $F^{-1}_{a,b}$ is the inverse CDF of a Beta distribution with parameters (a,b) (equivalent to the exact binomial tail inversion). For the SNR $= 25$ dB 4 B observation ($k = 5$, $n = 8.19 \times 10^5$ bits), this yields $[1.81 \times 10^{-6}, 1.30 \times 10^{-5}]$. For a zero-error observation ($k = 0$, $n = 8.19 \times 10^5$ bits, as in the 4 B PCSI+MMSE reference at SNR $= 25$ dB), the lower bound collapses to 0 and the upper bound is 4.5×10^{-6}; this is what we mean by "finite-sample censoring" in Chapters 3 and 4.

C OMP-Once Recall Against the Oracle Dirichlet Support (Probe (a))

Probe (a) of §3.4 is computed directly from the per-candidate binary labels already present in the feature cache produced by `afdm_sp_joint_unrolled/loop9_cache_features.py`, with no fresh Monte-Carlo required.

For each cached frame, the harvesting script runs one OMP-once pass to obtain the OMP-selected support S_{OMP}, computes the Dirichlet-weighted oracle top-P support S_{oracle} from the stored channel parameters, and assigns $y_i = 1$ iff candidate i is an oracle atom that OMP missed (i.e. cand_cols$_i \in S_{\text{oracle}} \setminus S_{\text{OMP}}$). The cache also stores $p_{\text{real}} = |S_{\text{OMP}}|$.

Because the OMP target sparsity equals the oracle top-P count in our configuration, $|S_{\text{OMP}}| = |S_{\text{oracle}}| = P$, so recall against the oracle support coincides with precision:

$$\text{recall} \;=\; \frac{|S_{\text{OMP}} \cap S_{\text{oracle}}|}{|S_{\text{oracle}}|} \;=\; \frac{p_{\text{real}} - \sum_i y_i}{p_{\text{real}}}.$$

The post-processing script `loop13_support_recall_probe.py` computes per-frame and per-SNR aggregates from this formula and writes `outputs_loop13_support_recall/recall_by_s`
The script runs in ≈ 30 s on the 27 k-frame 4 B cache; no GPU is required.

Precision as an independent quantity. Because $|S_{\text{OMP}}| = |S_{\text{oracle}}|$ in our configuration, precision is identically equal to recall on this dataset. Evaluating precision under a relaxed OMP sparsity (e.g. $|S_{\text{OMP}}| = P + 1$) would require replaying every frame from its stored seed to recover S_{OMP} explicitly; we leave that as a robustness follow-up and do not rely on it for any claim in this report.

D McNemar Paired-Bit Significance Test

On a paired bit-level comparison between the classical joint and the learned receiver, let n_{10} denote the number of bits on which the classical is wrong and the learned is correct, and n_{01} the number of bits on which the classical is correct and the learned is wrong. The McNemar [36] χ^2 statistic with continuity correction is

$$\chi^2_{\text{MN}} \;=\; \frac{(|n_{10} - n_{01}| - 1)^2}{n_{10} + n_{01}},$$

which is χ^2_1-distributed under the null hypothesis that the two receivers have the same bit-wise error probability. The one-sided p-value for the alternative "learned better" is $1 - F_{\chi^2_1}(\chi^2_{\text{MN}})$ when $n_{10} > n_{01}$, which is the number reported in Table 3. Two-sided p-values ($p \ll 10^{-4}$) are

invariant to the choice of side in our regime because n_{10} dominates n_{01} by multiple orders of magnitude at the two high-SNR points where the test is applied.